1.1 OUTPUT

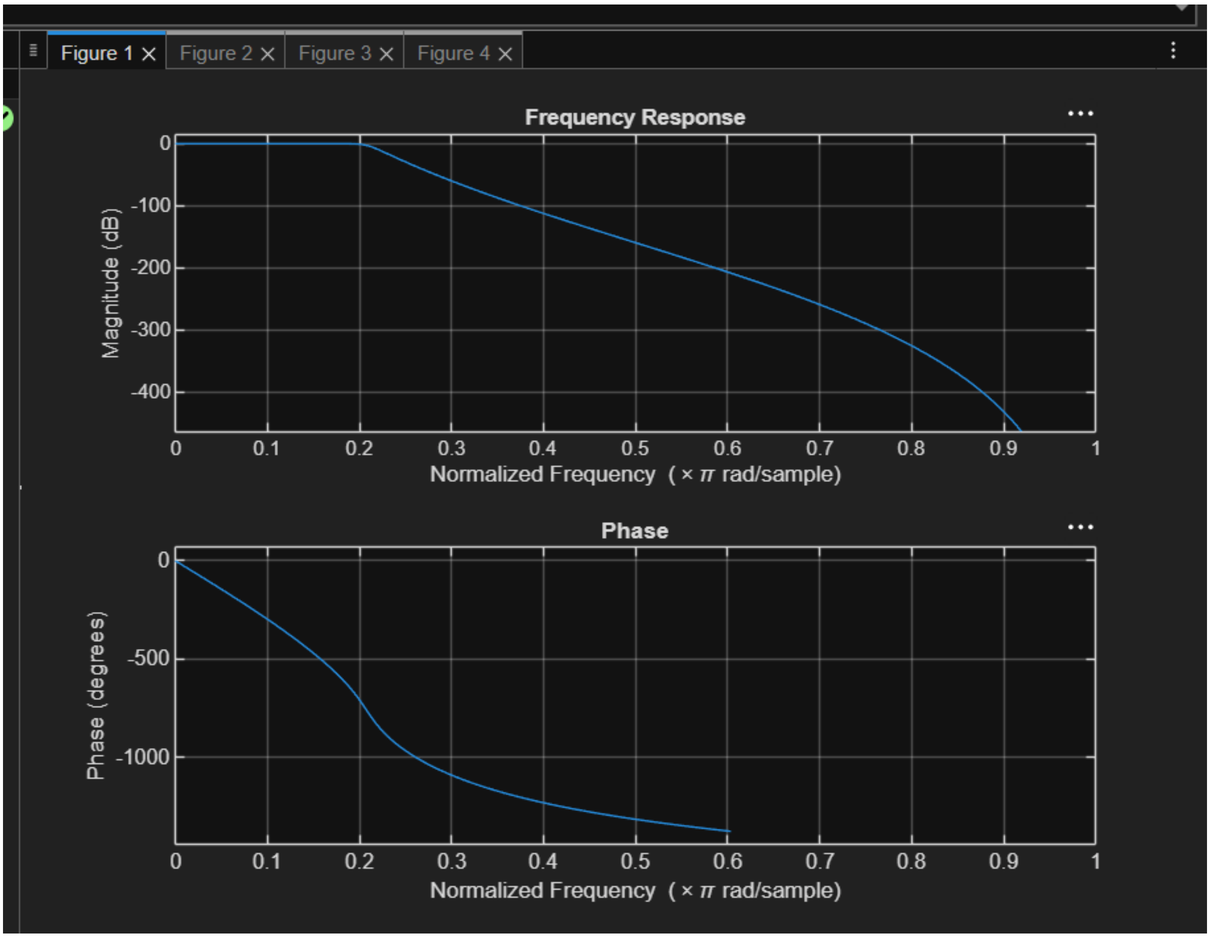
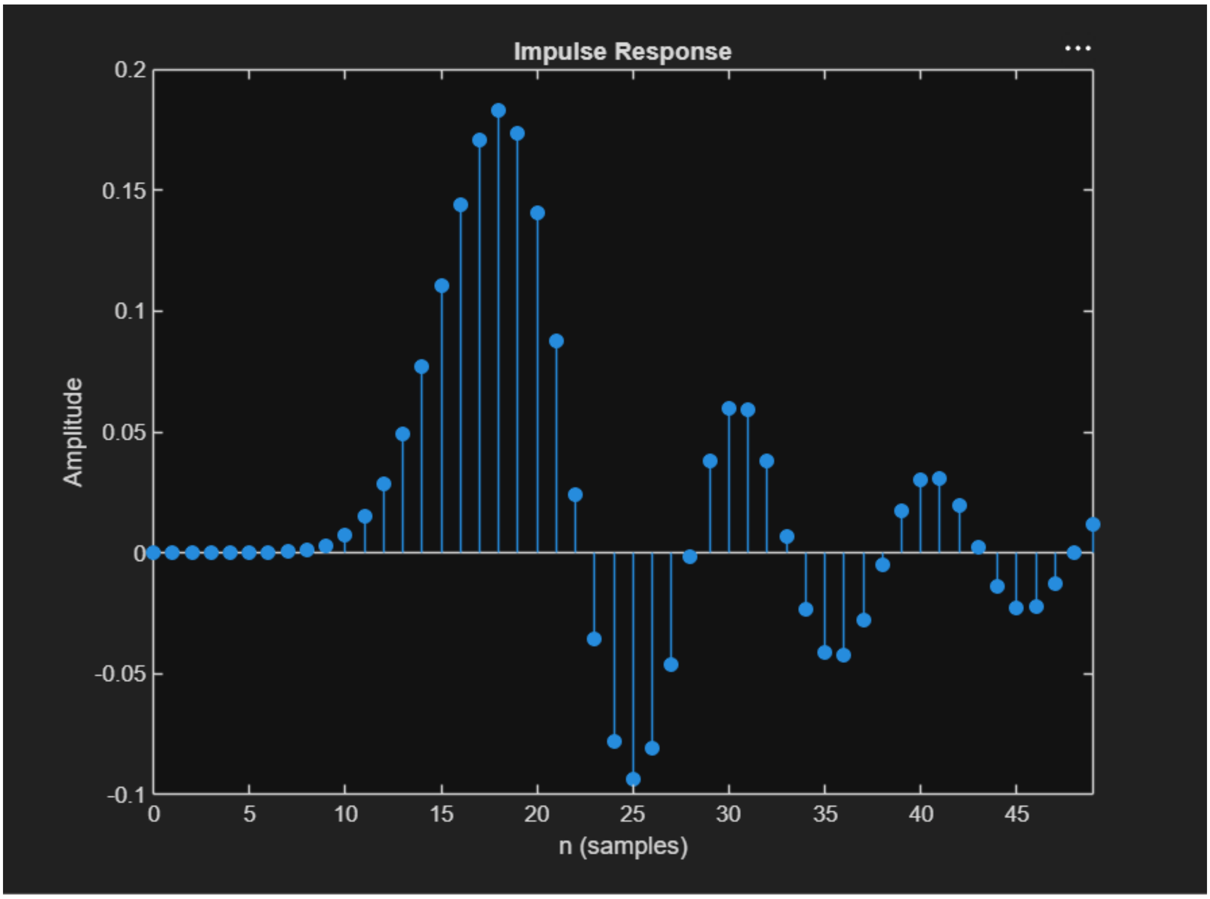
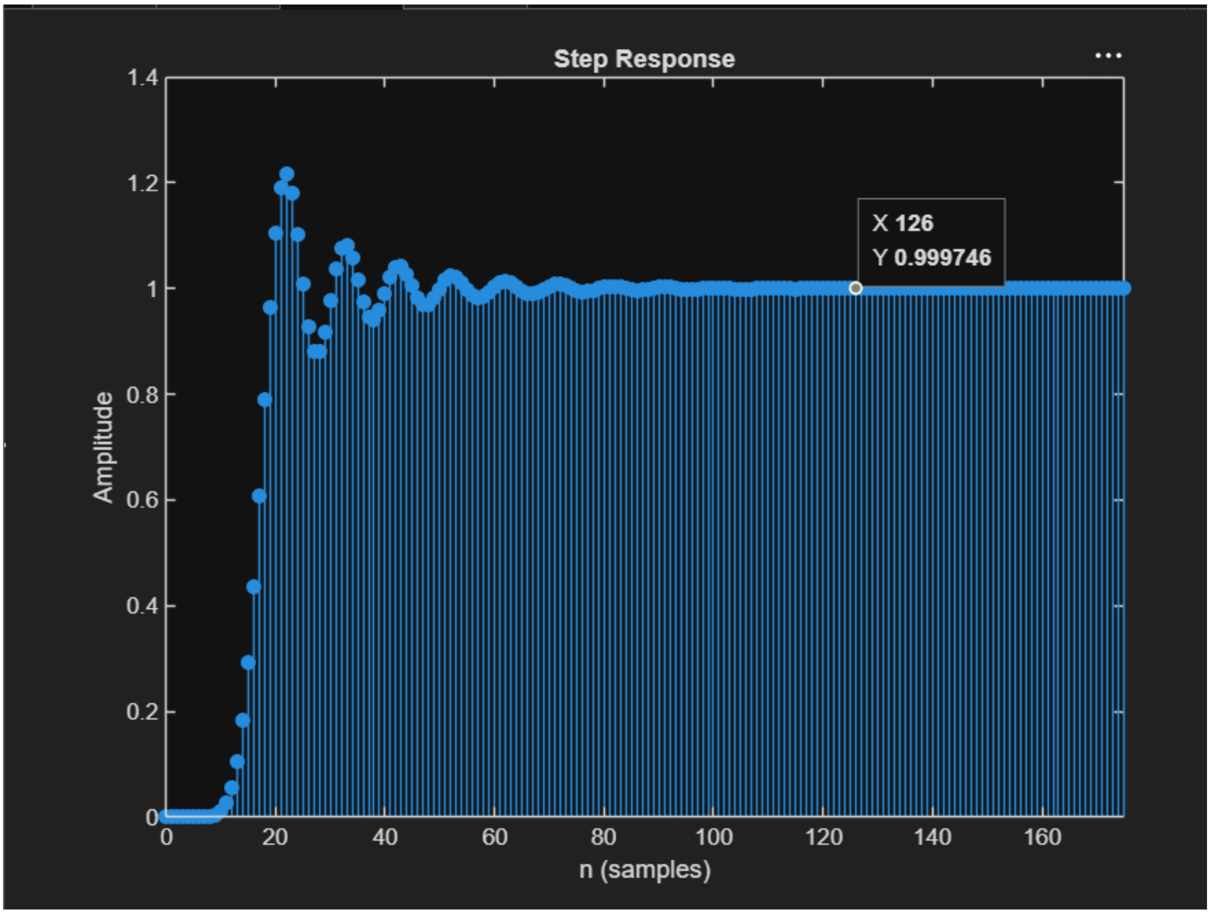
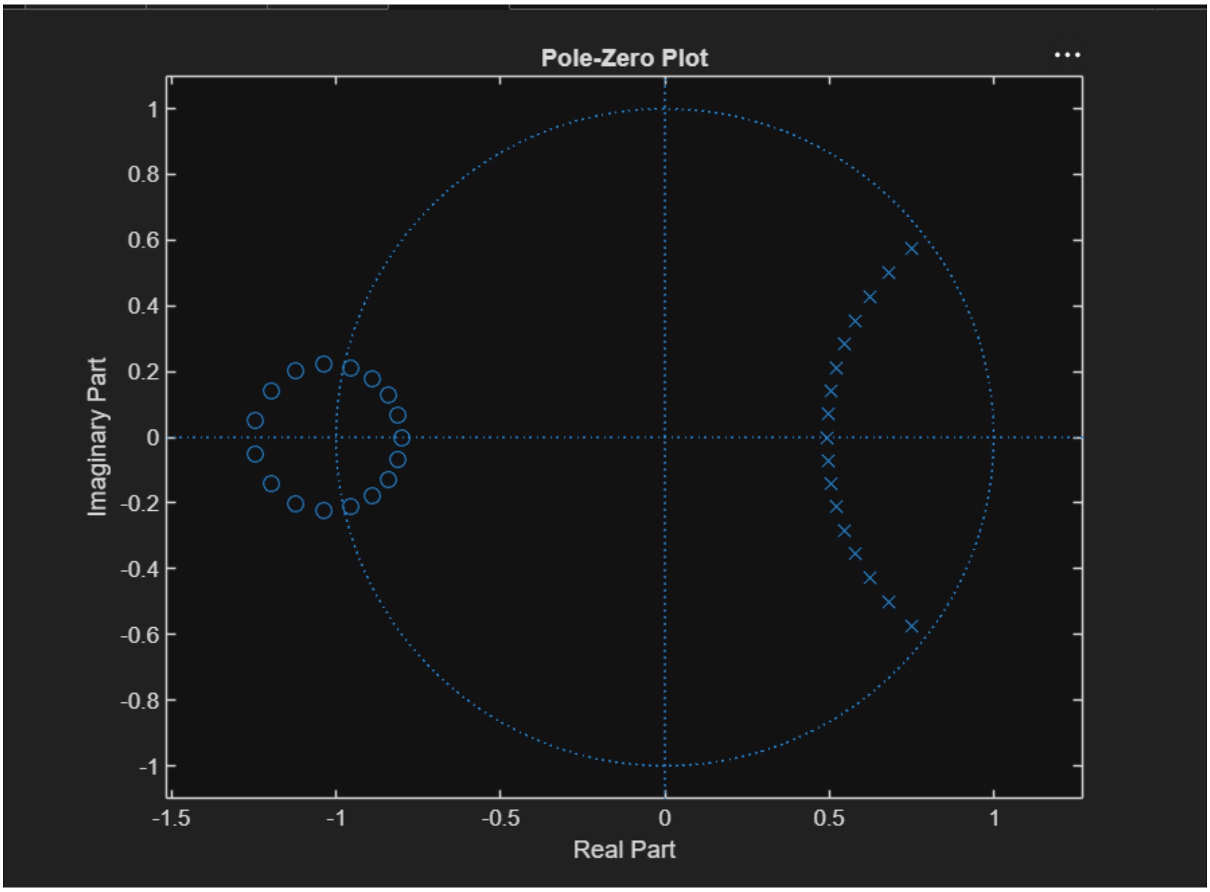


Figure 1
Figure 2
Figure 3
Figure 4
Frequency Response
Magnitude (dB)
0
-100
-200
-300
-400
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)


Impulse Response
Amplitude
n (samples)
0.2
0.15
0.1
0.05
0
-0.05
-0.1
0
5
10
15
20
25
30
35
40
45


Step Response
X 126
Y 0.999746
Amplitude
n (samples)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
20
40
60
80
100
120
140
160



PROGRAM

```
clc;
clear;
close all;

% EX NO: 1 Butterworth IIR Filters
% Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB

Fs = 10000;
Fp = 1000;
Fst = 1500;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a] = butter(N,Wn);
```

```matlab
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');

% Impulse Response
figure;
impz(b,a,50);
title('Impulse Response');

% Step Response
figure;
stepz(b,a);
title('Step Response');

% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```